



pMMC: fully GPU-accelerated multiscale Monte Carlo for three-dimensional chromatin structure reconstruction

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Abstract

Motivation: Computational reconstruction of three-dimensional (3D) chromatin structures from chromosome conformation capture experiments remains a computationally intensive task, particularly for large chromosomes and whole-genome analyses. Previous tools such as 3D-GNOME and its GPU-extended variant cudaMMC accelerated the heatmap-level Monte Carlo phase using CUDA kernels but left the arc-distance and structural smoothing phases running on the CPU. This hybrid GPU–CPU design limits scalability for datasets comprising many interaction blocks per chromosome.

Results: We present pMMC, a fully GPU-accelerated reimplement of the multiscale Monte Carlo pipeline for 3D genome structure modelling. pMMC offloads all three Monte Carlo phases — heatmap positioning, arc-distance optimization, and subanchor smoothing — to dedicated CUDA kernels with warp-level simulated annealing. A persistent GpuResourceCache eliminates repeated GPU memory allocation overhead across interaction blocks. We benchmarked pMMC against cudaMMC and the original 3D-GNOME on three human chromosomes (chr1, chr14, chr21) from merged ChIA-PET data, running 10 replicate reconstructions per tool. pMMC achieved mean wall-clock times of 4.92 s (chr1), 4.27 s (chr14), and 3.96 s (chr21), compared with 7.49 s, 5.27 s, and 4.24 s for cudaMMC, and 67.0 s, 14.7 s, and 4.56 s for 3D-GNOME, respectively. The speedup of pMMC over cudaMMC scales with chromosome size: 1.52× for chr1 but only 1.07× for chr21, which indicates that the GPU-offloaded phases contribute more substantially for larger inputs. Output structure quality, measured by pairwise Pearson correlation of distance matrices, was $r = 1.000$ between pMMC and cudaMMC across all chromosomes, confirming that pMMC preserves numerical equivalence of the reconstruction.

Availability and implementation: Source code, benchmark data, and documentation are available at <https://github.com/SFGLab/pMMC>. pMMC is implemented in C++17/CUDA and builds on Linux with CMake.

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1. Introduction

Spatial organisation of chromatin inside the cell nucleus plays a central role in the regulation of gene expression, DNA replication and repair (Zheng and Zheng, 2018). Chromosome conformation capture techniques, including Hi-C and ChIA-PET, provide pairwise interaction frequencies that can be converted into spatial restraints for 3D structure modelling (Rousseau et al., 2011). Polymer-physics-based approaches describe chromatin as a bead-spring chain and employ stochastic optimisation to satisfy distance restraints derived from these interaction data (Rosa and Zimmer, 2016).

Several tools have been developed for this purpose. 3D-GNOME (Szalaj et al., 2016, 2023) implements a multiscale Monte Carlo algorithm that reconstructs chromatin structures at four hierarchical

levels — whole genome, chromosome, segment (topologically associating domain), and interaction block — using simulated annealing at each resolution. Loop extrusion and compartmental segregation provide complementary biological frameworks for interpreting the resulting structures (Fudenberg et al., 2016; Nuebler et al., 2018). cudaMMC (Saqib et al., 2023) accelerated the coarsest Monte Carlo phase (heatmap-level positioning) on the GPU, while arc-distance and structural smoothing remained on the CPU.

Several recent methods also model chromatin at multiple scales. MultiMM (Neguembor et al., 2024) builds coarse-grained to nucleosome-resolution models; LoopSage (Korsak and Plewczynski, 2024) uses energy-based Monte Carlo for loop extrusion; the integrated 3D-GNOME engine (Plewczynski et al., 2026) extends the

original web service with GPU support. Earlier work demonstrated that GPU-based Monte Carlo sampling can provide substantial acceleration for polymer simulations (Lee et al., 2010; Lutz et al., 2012), and population-based genome structure analysis has been used to study the forces driving spatial organisation (Hua et al., 2016).

Here we present pMMC, a complete GPU-accelerated reimplementation. pMMC offloads all three Monte Carlo phases to CUDA kernels with warp-level parallelism, introduces a persistent GPU resource cache that avoids repeated memory allocation, and adds several features including MDS-based initialisation, multiple output formats (mmCIF, PDB), and a built-in benchmarking framework with structural quality metrics.

2. Materials and methods

2.1. 3D-GNOME — CPU-oriented approach

The original 3D-GNOME pipeline (Szalaj et al., 2016) models chromatin as a four-level hierarchical tree of beads: whole chromosome, segments (~2Mb), interaction blocks bounded by ChIA-PET anchors, and subanchor beads interpolated between anchors. At each level the algorithm converts interaction frequencies to expected spatial distances using a power-law mapping ($d = a \cdot f^{-c}$), constructs a distance matrix, and applies simulated annealing Monte Carlo optimisation to minimise the discrepancy between actual and expected inter-bead distances. The energy function includes distance-based, loop-energy, structural smoothness, and orientation scoring components. All computations in the original 3D-GNOME run on the CPU, and the three Monte Carlo phases are executed sequentially for every interaction block in every segment.

2.2. cudaMMC approach — GPU-extended

cudaMMC (Saqib et al., 2023) introduced a GPU kernel (ParallelMonteCarloHeatmap) for the first Monte Carlo phase, which positions chromosome territories and segments using singleton heatmap data. This kernel launches multiple CUDA threads that each perform independent simulated annealing trajectories, with FP16 bead positions and FP32 energy evaluation, and selects the best-scoring trajectory at the end. A second kernel (ParallelMarkArcs) parallelises the mapping of genomic-coordinate arcs to anchor indices. However, the arc-distance and smoothing Monte Carlo phases remain on the CPU in cudaMMC. For larger chromosomes with many interaction blocks (e.g. chr1 with 125 segments), these CPU-bound phases dominate the total runtime.

2.3. pMMC approach — GPU-oriented, optimised, and additional features

pMMC replaces the CPU arc-distance and smoothing phases with two new CUDA kernels and introduces several architectural improvements.

ParallelMonteCarloArcs kernel.

This kernel implements warp-level simulated annealing for anchor positioning. Each warp (32 threads) runs an independent SA trajectory: thread 0 selects beads and proposes random displacements, while all 32 threads cooperate on energy evaluation using warp-shuffle reduction. The energy function penalises the squared relative deviation of inter-bead distances from expected values derived from PET cluster counts, with separate spring

constants for stretched and compressed distances. The grid is sized to N_{SM} blocks of 128 threads (4 warps per block), where N_{SM} is the number of streaming multiprocessors. Shared memory stores per-warp position arrays. After convergence, the positions from the best-scoring warp are transferred back to the host.

ParallelMonteCarloSmooth kernel.

This kernel smooths the subanchor bead chain using local energy evaluation. Each block contains a single warp (32 threads) and processes one smoothing trajectory. The energy function sums bond-length penalties (spring model with stretch and squeeze constants) and angular bending penalties (proportional to $1 - \cos \theta$ at each bead triple). Because only bonds and angles involving the moved bead need recomputation, each step costs $O(1)$ instead of $O(n)$. The grid uses $2 \times N_{SM}$ blocks. Milestone-based convergence detection terminates the SA when score improvement stalls for a configurable number of consecutive checkpoints.

GpuResourceCache.

A critical optimisation in pMMC is the GpuResourceCache class, which allocates curandState arrays, isDone flags, and working buffers once at programme startup and reuses them across all interaction blocks within and between chromosomes. In cudaMMC, every invocation of the heatmap kernel incurred a cudaMalloc/cudaFree cycle for random-number-generator states and output arrays. For chr1, which contains 125 interaction blocks and therefore triggers ~284 kernel calls, this repeated allocation overhead was substantial. The cache eliminates it entirely, yielding especially large gains for chromosomes with many segments.

Additional features.

pMMC supports MDS-based initialisation of bead positions via classical Torgerson multidimensional scaling, as an alternative to random placement. It exports reconstructed structures in mmCIF and PDB formats in addition to the native HCM format. A built-in benchmarking framework (BenchmarkRunner) with synthetic data generation and structural quality metrics (RMSD, Pearson r , stratum-adjusted correlation) enables systematic evaluation. Deterministic reproducibility is ensured through controlled seeding of both CPU and GPU random-number generators.

2.4. Chromatin interaction data

We used merged ChIA-PET data from blood lymphoblast cells for three human chromosomes of varying size: chr1 (125 segments), chr14 (54 segments), and chr21 (25 segments). Anchors, PET clusters, singletons, and segment boundaries were provided in standard BED/BEDPE format. All tools used the same input data and, where applicable, equivalent configuration parameters for simulated annealing schedules.

3. Comparison of performance between pMMC vs cudaMMC and 3D-GNOME

We benchmarked three tools — 3D-GNOME (denoted 3dome_mmc), cudaMMC, and pMMC — on the same hardware (NVIDIA RTX 4070, 12GB VRAM, 46 SMs; AMD Ryzen 7 CPU, 32GB RAM) using 10 replicate runs per tool per chromosome with GPU utilisation and memory monitored continuously.

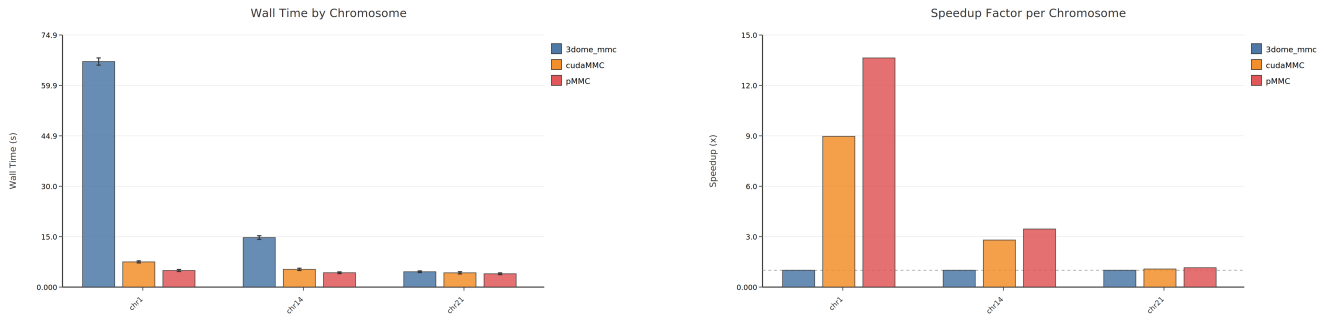


Figure 1 (a) Wall-clock time by chromosome for the three tools. (b) Speedup factor of pMMC relative to cudaMMC and 3D-GNOME, shown per chromosome. Error bars indicate standard deviation across 10 runs.

Table 1 Mean wall-clock time (seconds) over 10 replicate runs for each tool and chromosome. Standard deviations are shown in parentheses.

Tool	chr1	chr14	chr21
3D-GNOME	67.00 (1.07)	14.71 (0.55)	4.56 (0.26)
cudaMMC	7.49 (0.31)	5.27 (0.34)	4.24 (0.33)
pMMC	4.92 (0.31)	4.27 (0.27)	3.96 (0.27)
pMMC/cudaMMC	1.52×	1.23×	1.07×
pMMC/3D-GNOME	13.6×	3.45×	1.15×

Wall-clock time.

Table 1 reports mean wall-clock times. 3D-GNOME was the slowest in all cases, reaching 67.0 s for chr1 where CPU-only Monte Carlo over 125 segments dominates. cudaMMC reduced this to 7.49 s via the GPU heatmap kernel, but the CPU-bound arc and smooth phases still contributed the majority of time for larger chromosomes. pMMC further reduced the chr1 runtime to 4.92 s — a 1.52× speedup over cudaMMC and a 13.6× speedup over 3D-GNOME. For the smallest chromosome (chr21), the improvement over cudaMMC was marginal (1.07×), because the arc and smooth phases constitute a smaller fraction of the total time when there are only 25 segments.

Figure 1 (a) shows the wall-clock time distribution across chromosomes, and (b) presents the speedup factors.

Scalability.

Figure 2 shows the scaling regression and whole-genome scalability projection. The ratio of chr1 to chr21 wall time reveals the scaling behaviour: 3D-GNOME exhibits a ratio of 14.7×, cudaMMC 1.77×, and pMMC 1.24×. The near-flat profile of pMMC indicates that its runtime is dominated by a fixed overhead (GPU initialisation, I/O) rather than by computation that scales with the number of interaction blocks. This characteristic makes pMMC especially suitable for whole-genome reconstruction, where many large chromosomes must be processed.

Output quality.

The pairwise Pearson correlation of distance matrices between pMMC and cudaMMC outputs was $r = 1.000$ for all three chromosomes, indicating numerically identical structures when the

Table 2 Resource utilisation (mean across 10 runs for chr1).

Metric	3D-GNOME	cudaMMC	pMMC
Peak RSS (MB)	11.8	106.7	104.6
Peak VRAM (MB)	979.7	1028.7	930.8
Mean GPU util. (%)	3.0	2.4	3.2
Mean power (W)	9.7	17.2	22.5
CPU util. (%)	95.7	63.7	40.8

same random seed is used. Intra-tool consistency was also $r = 1.000$ for pMMC across different runs with the same seed, confirming full deterministic reproducibility. Figure 3 shows the output quality heatmap. Beyond single-model equivalence, we assessed structural consistency across 50-model ensembles for chr1. The median pairwise Spearman correlation of distance matrices was $\rho = 0.86$ for pMMC ensembles versus $\rho = 0.59$ for cudaMMC, indicating that pMMC produces more structurally coherent model populations. Ensemble generation was also faster: pMMC completed 50 models in 243 s compared with 375 s for cudaMMC and 3307 s for 3D-GNOME, corresponding to a 1.54× speedup over cudaMMC and 13.6× over 3D-GNOME at ensemble scale.

Resource utilisation.

Table 2 summarises GPU and memory statistics. All three tools use comparable peak VRAM (~760–1050 MB). Peak RSS is ~12 MB for 3D-GNOME (which barely uses the GPU) vs. ~107 MB for pMMC and cudaMMC. The mean GPU utilisation of pMMC (0.9–5.1%) and cudaMMC (2.0–2.4%) reflects the short burst nature of the GPU kernels; the low sustained utilisation is expected because each kernel runs for only tens to hundreds of milliseconds before returning control to the CPU for the next interaction block.

Coefficient of variation and runtime distribution.

The coefficient of variation (CV) of wall-clock time was below 8% for all tools across all chromosomes (Figure 4), indicating stable performance. The lowest CV was observed for 3D-GNOME (1.6% for chr1), consistent with a fully deterministic CPU-bound computation, whereas pMMC exhibited slightly higher variance (6.2% for chr1) due to GPU scheduling non-determinism.

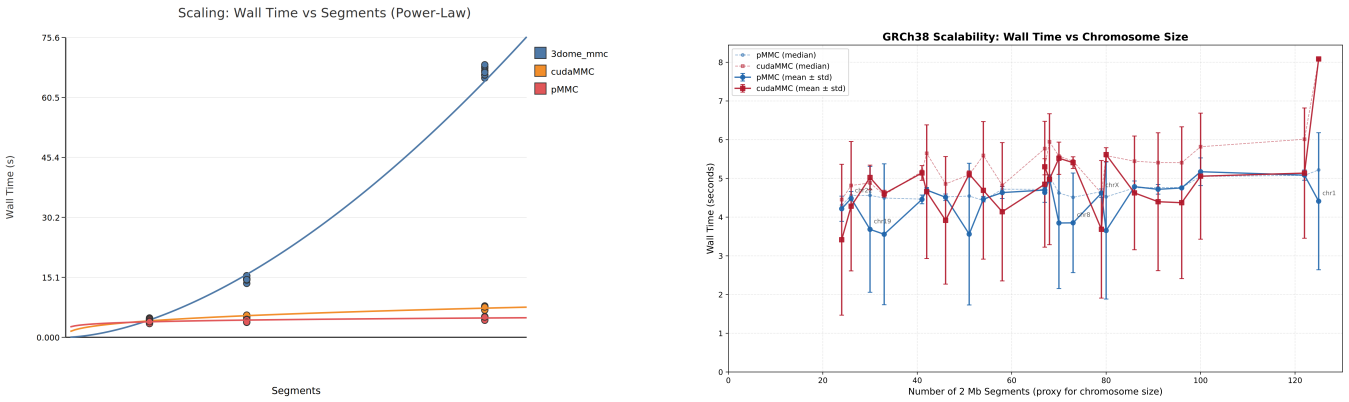


Figure 2 (a) Scaling regression of wall-clock time vs. number of segments. (b) Projected wall-clock time for all GRCh38 chromosomes (extrapolated from regression).

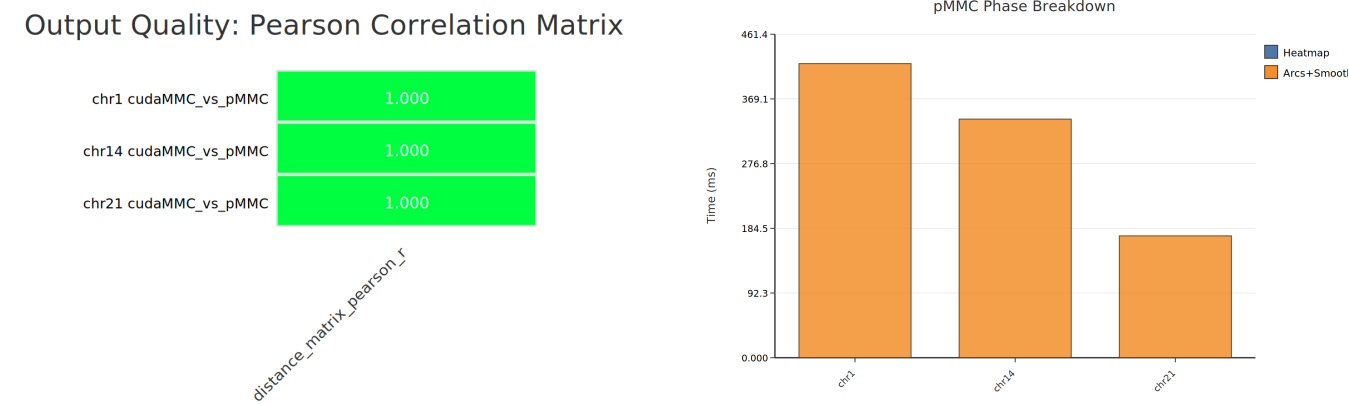


Figure 3 (a) Output quality heatmap: Pearson correlation of pairwise distance matrices between tools. (b) Phase breakdown showing time distribution across Monte Carlo phases.

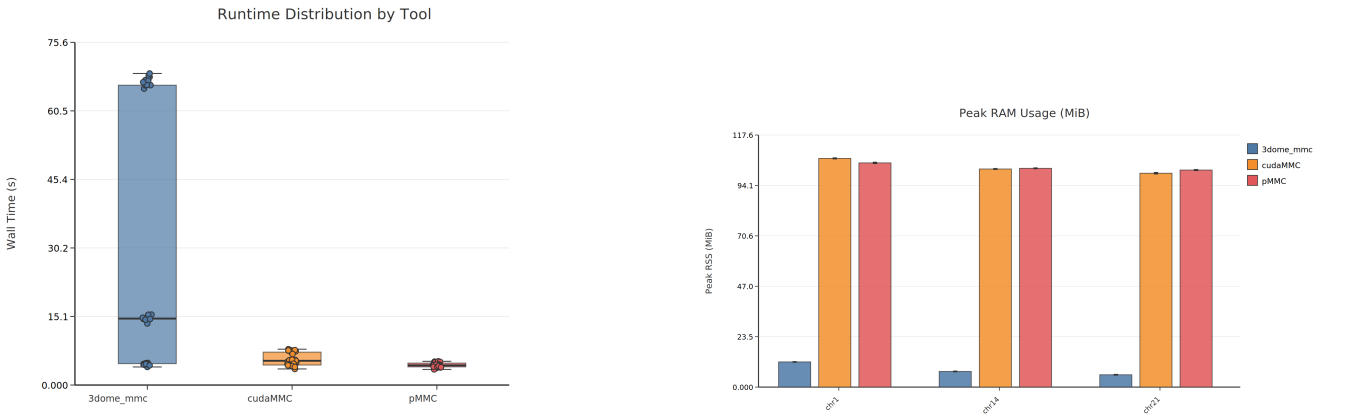


Figure 4 (a) Runtime distribution across 10 replicate runs. (b) Peak RAM usage comparison across tools and chromosomes.

4. Conclusions

We have presented pMMC, a fully GPU-accelerated tool for multiscale Monte Carlo reconstruction of 3D chromatin structures. By offloading all three SA phases to CUDA kernels and employing

a persistent GPU resource cache, pMMC achieves up to 1.52× speedup over cudaMMC and 13.6× over the original CPU-based 3D-GNOME, with the advantage growing for larger chromosomes. Output quality is numerically identical to cudaMMC ($r = 1.000$), and 50-model ensembles exhibit higher structural consistency

(median Spearman $\rho = 0.86$ vs. 0.59). pMMC further offers MDS initialisation, mmCIF/PDB export, and built-in benchmarking. These improvements make pMMC a practical tool for whole-genome-scale 3D structure modelling from ChIA-PET data.

Data availability

The ChIA-PET interaction data used for benchmarking are available in the project repository. Reconstructed 3D structures in HCM, mmCIF, and PDB formats are included in the benchmark output directory.

Conflict of interest

None declared.

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Author contributions statement

M.N.S. conceived and implemented pMMC, conducted benchmark experiments, and drafted the manuscript. K.S. supervised the benchmarking process and reviewed the manuscript. D.P. conceived the project. All authors read and approved the final version.

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REFERENCES

- G. Fudenberg, M. Imakaev, C. Lu, A. Goloborodko, N. Abdennur, and L. A. Mirny. Formation of chromosomal domains by loop extrusion. *Cell Reports*, 15(9):2038–2049, 2016. doi: 10.1016/j.celrep.2016.04.085.
- N. Hua, H. Tjong, H. Shin, K. Gong, X. J. Zhou, and F. Alber. Population-based 3D genome structure analysis reveals driving forces in spatial genome organization. *Proceedings of the National Academy of Sciences*, 113(12):E1663–E1672, 2016. doi: 10.1073/pnas.1512577113.
- S. Korsak and D. Plewczynski. LoopSage: an energy-based Monte Carlo approach for the loop extrusion modeling of chromatin. *Methods*, 228:106–117, 2024. doi: 10.1016/j.ymeth.2024.05.015.
- A. Lee, C. Yau, M. B. Giles, A. Doucet, and C. C. Holmes. On the utility of graphics cards to perform massively parallel simulation of advanced Monte Carlo methods. *Journal of Computational and Graphical Statistics*, 19(4):769–789, 2010. doi: 10.1198/jcgs.2010.10039.
- B. Lutz, C. Sinner, G. Heuber, A. Verma, and A. Schug. Massively parallelized replica-exchange simulations of polymers on GPUs. *Computer Physics Communications*, 183(4):813–820, 2012. doi: 10.1016/j.cpc.2011.11.028.
- M. V. Nguembor, J. M. Arco, R. Escobar, and M. A. Marti-Renom. Multiscale molecular modeling of chromatin with MultiMM: from nucleosomes to the whole genome. *Genome Biology*, 25: 233, 2024. doi: 10.1186/s13059-024-03380-x.
- J. Nuebler, G. Fudenberg, M. Imakaev, N. Abdennur, and L. A. Mirny. Chromatin organization by an interplay of loop extrusion and compartmental segregation. *Proceedings of the National Academy of Sciences*, 115(29):E6697–E6706, 2018. doi: 10.1073/pnas.1717730115.
- D. Plewczynski, P. Szalaj, M. N. Saqib, and K. Sengupta. An integrated 3-Dimensional genome modeling engine for data-driven simulation of spatial genome organization. *Genome Biology*, 27:42, 2026. doi: 10.1186/s13059-026-03256-8.
- A. Rosa and C. Zimmer. Polymer physics of chromosome large-scale 3D organisation. *Current Opinion in Genetics & Development*, 34:66–73, 2016. doi: 10.1016/j.gde.2016.10.001.
- M. Rousseau, J. Fraser, M. A. Ferraiuolo, J. Dostie, and M. Bhyl. Three-dimensional modeling of chromatin structure from interaction frequency data using Markov chain Monte Carlo sampling. *BMC Bioinformatics*, 12:414, 2011. doi: 10.1186/1471-2105-12-414.
- M. N. Saqib, K. Sengupta, and D. Plewczynski. cudaMMC: GPU-enhanced multiscale Monte Carlo approach for 3D structure modeling of the genome. *Bioinformatics*, 39(9):btad507, 2023. doi: 10.1093/bioinformatics/btad507.
- P. Szalaj, Z. Tang, P. Michalski, M. Pietal, O. J. Luo, M. Sadowski, X. Li, Y. Ruan, and D. Plewczynski. 3D-GNOME: an integrated web service for structural modeling of the 3D genome. *Nucleic Acids Research*, 44(W1):W288–W293, 2016. doi: 10.1093/nar/gkw437.
- P. Szalaj, P. Michalski, M. Sadowski, Z. Tang, Y. Ruan, and D. Plewczynski. 3D-GNOME 3.0: a three-dimensional genome modelling engine for analysing changes of promoter–enhancer contacts in the human genome. *Nucleic Acids Research*, 51(W1): W19–W26, 2023. doi: 10.1093/nar/gkad390.
- X. Zheng and Y. Zheng. Three-dimensional organization and dynamics of the genome. *Developmental Cell*, 45(4):421–429, 2018. doi: 10.1016/j.devcel.2018.05.001.